SHORE

Name:

Set: SOLUTIONS

Year 10

Standard Mathematics

14th March 2025

Task 1: Sets 1-3

General Instructions

- Reading time – 0 minutes
- Working time – 45 minutes
- Write using black or blue non-erasable pen.
- Draw diagrams using pencil.
- Calculators may be used.
- Marks may not be awarded for untidy or badly arranged work.
- Label each page with your name and set.

Notes:

- Diagrams are NOT drawn to scale unless stated otherwise.
- Any time you have remaining should be spent revising your answers.

DO NOT REMOVE THIS PAPER FROM THE EXAMINATION ROOM

		Marks Allocated	Marks Awarded
Booklet A	Area and Surface Area	**16**	
Booklet B	Linear Relationships	**26**	
	TOTAL	**42**	

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2025

Mathematics Standard 1 – 3

Answer Booklet

A

Booklet A

16 marks

Attempt Questions 1 – 8

Allow about 18 minutes for this Booklet

Area and Surface Area

Instructions

- Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response.
- Your responses should include relevant mathematical reasoning and/or calculations.
- Extra writing space is provided on the back of the booklet. If you use this space, clearly indicate which question you are answering.

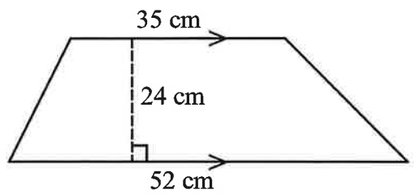
Please turn over

Booklet A: Area and Surface Area **16 Marks**

Write all necessary working and your answers in the spaces provided.

Multiple Choice for questions 1 to 3.
Circle the correct answer.

1 The following shape is a trapezium. **1**



NOT TO SCALE

Which of the following is the area of the shape?

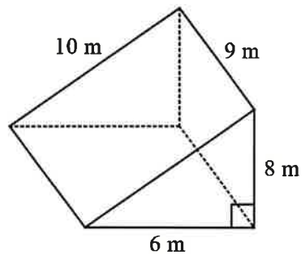
A. 612 cm^2

B. 1044 cm^2

C. 1131 cm^2

D. 2088 cm^2

2 The following shape is a triangular prism. **1**



NOT TO SCALE

Which of the following is the triangular prism's surface area?

A. 24 m^2

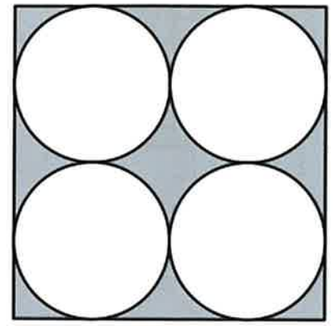
B. 258 m^2

C. 264 m^2

D. 312 m^2

/2

3 In the diagram below, each of the circles has a diameter of 6 cm and the edges of the square touch the circumference of each circle. **1**



NOT TO SCALE

Which of the following is the area of the shaded region to the nearest square centimetre?

A. 31 cm^2

B. 69 cm^2

C. 77 cm^2

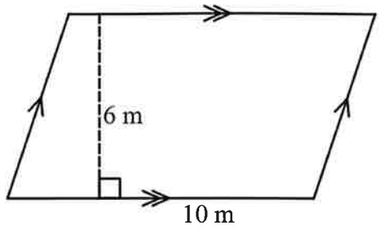
D. 89 cm^2

/1

Question 4 (3 marks)

Calculate the area of the following shapes.

(a) **1**



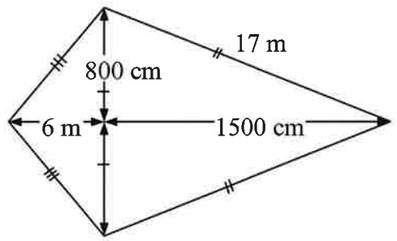
NOT TO SCALE

$A = bh$

$= 6 \times 10$

$= 60 m^2$

(b) **2**



NOT TO SCALE

$A = \frac{1}{2} xy$

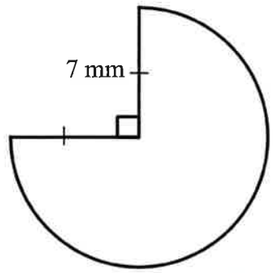
$= \frac{1}{2}(8+8)(6+15)$

$= 168 m^2$ or $1\,680\,000 cm^2$

/3

Question 5 (2 marks)

Calculate the perimeter of the following sector, leaving your answer correct to 2 decimal places. **2**



NOT TO SCALE

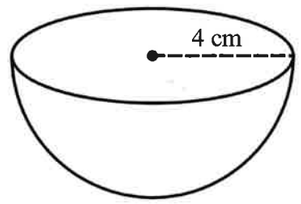
$P = 7 + 7 + \frac{3}{4} \times 2 \times \pi \times 7$

$= 46.986...$

$= 46.99$ mm

Question 6 (2 marks)

A bowl is in the shape of a **closed** hemisphere with a radius of 4 cm.



NOT TO SCALE

What is the surface area of the bowl? **2**

Give your answer correct to the nearest square centimetre.

$SA = 3\pi r^2$

$= 3\pi(4)^2$

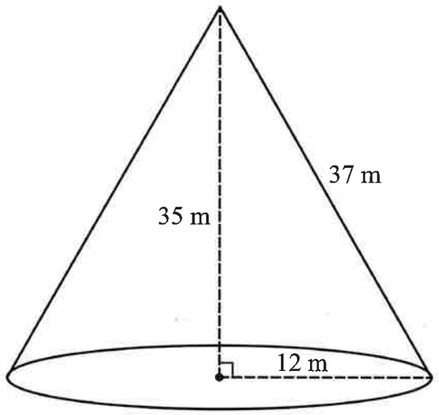
$= 150.79...$

$= 151 cm^2$

/4

Question 7 (2 marks)

Consider the closed cone below.



NOT TO SCALE

What is the **exact** surface area of the **closed** cone? **2**

$SA = \pi r^2 + \pi r l$

$= \pi (12)^2 + \pi (12)(37)$

$= 144\pi + 444\pi$

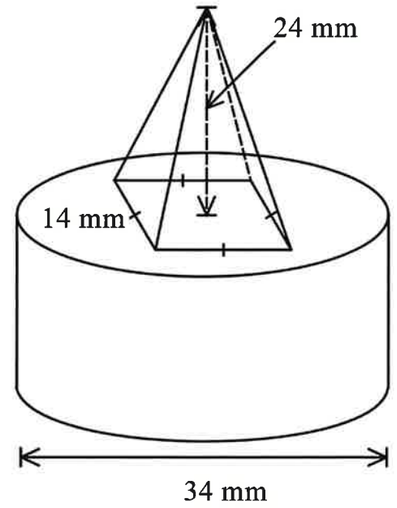
$= 588\pi \text{ m}^2$

/2

Question 8 (4 marks)

The composite solid below is formed by mounting a square pyramid on top of a cylinder.

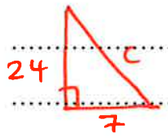
The perpendicular height of the pyramid is 24 mm.



NOT TO SCALE

The outside of the shape is to be painted. What is the total area covered? **4**

Give your answer to the nearest square millimetre.



$c^2 = 24^2 + 7^2$

$= 625$

$c = 25$

$SA = 2\pi r^2 + 2\pi rh - s^2 + 4sc \times \frac{1}{2}$

where r is the radius of the cylinder

h is the height of the cylinder

s is the sides of the square base of the pyramid

$\Rightarrow SA = 2\pi(17)^2 + 2\pi(17)(15) - 14^2 + 4(14)(25) \times \frac{1}{2}$

$\doteq 3922.05\ldots$

$\doteq 3922 \text{ mm}^2$

END OF BOOKLET A

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2025

Name:

Set: SOLUTIONS

Mathematics Standard 1-3

Answer Booklet

B

Booklet B

26 marks

Attempt Questions 9 – 18

Allow about 27 minutes for this booklet

Linear Relationships

Instructions

- Write your Name and Set at the top of this page.
- Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response.
- Your responses should include relevant mathematical reasoning and/or calculations.
- Extra writing space is provided on the back of the booklet. If you use this space, clearly indicate which question you are answering.

Please turn over

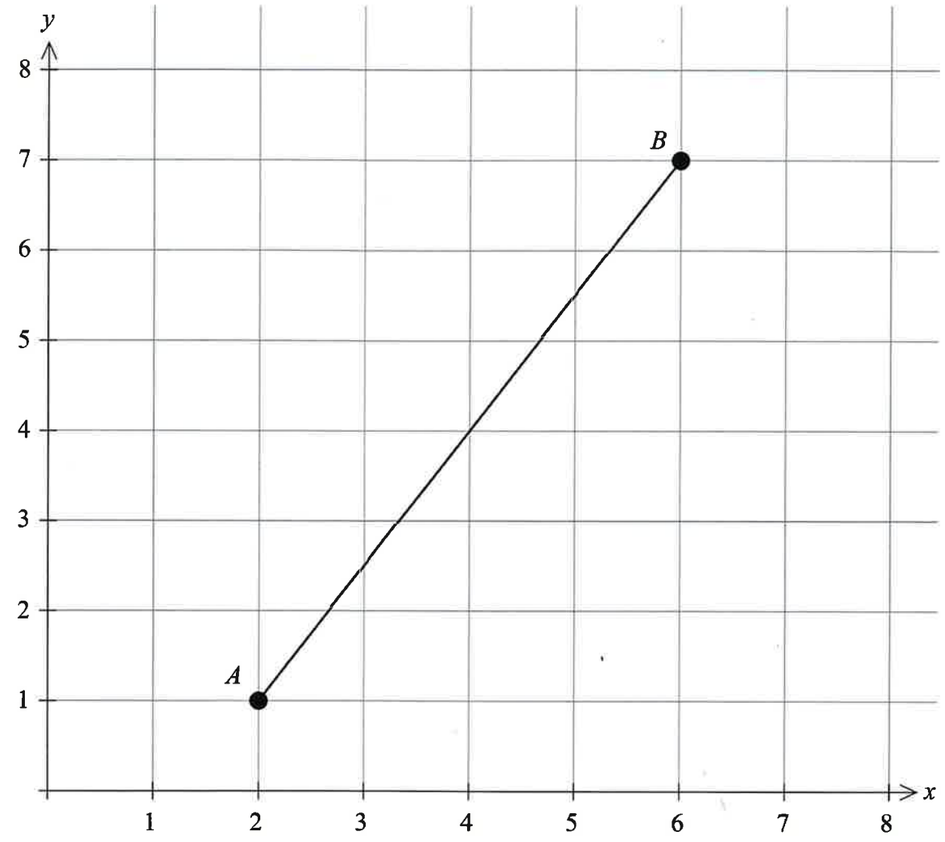
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Booklet B: Linear Relationships **26 Marks**

Write all necessary working and your answers in the spaces provided.

Multiple Choice for questions 9 to 11.
Circle the correct answer.

9 The following cartesian plane plots an interval AB.



What is the gradient of interval AB? **1**

A. $\frac{2}{3}$

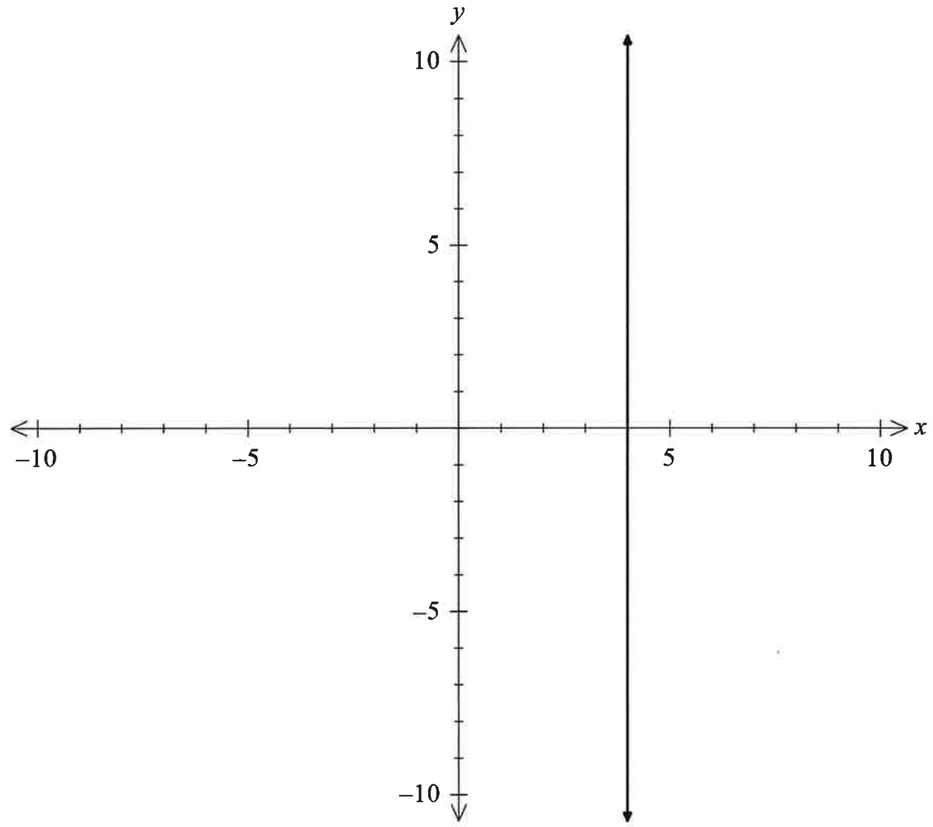
B. $\frac{6}{7}$

C. $\frac{7}{6}$

D. $\frac{3}{2}$

/1

10 Consider the following graph.



Which of the following is the correct equation that represents the graph shown? **1**

A. $y = x + 4$

B. $y = 4$

C. $x + y = 4$

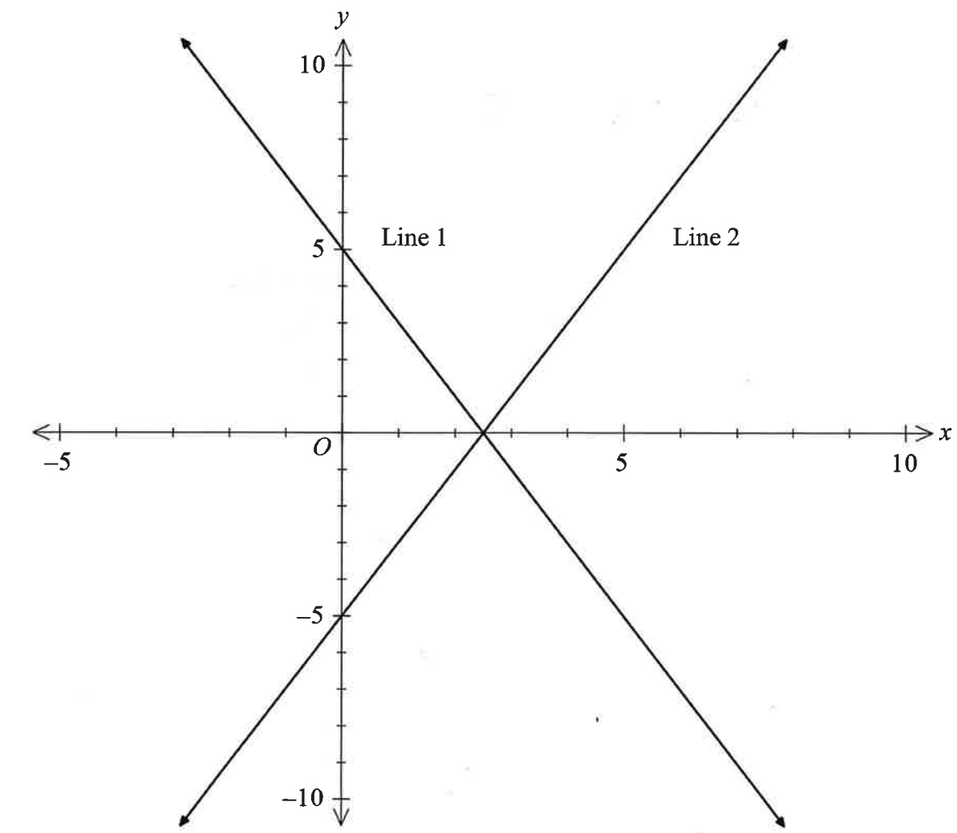
D. $x = 4$

/1

11 A computer application was used to draw the graphs of the equations:

$$y = -2x + 5 \text{ and } y = 2x - 5.$$

Part of the screen is shown.



Which row correctly matches the equations with the lines drawn and identifies the point of intersection? **1**

	$y = -2x + 5$	$y = 2x - 5$	Solution
A.	Line 1	Line 2	$x = 2.5,\ y = 0$
B.	Line 1	Line 2	$x = 5,\ y = -5$
C.	Line 2	Line 1	$x = 2.5,\ y = 0$
D.	Line 2	Line 1	$x = 5,\ y = -5$

/1

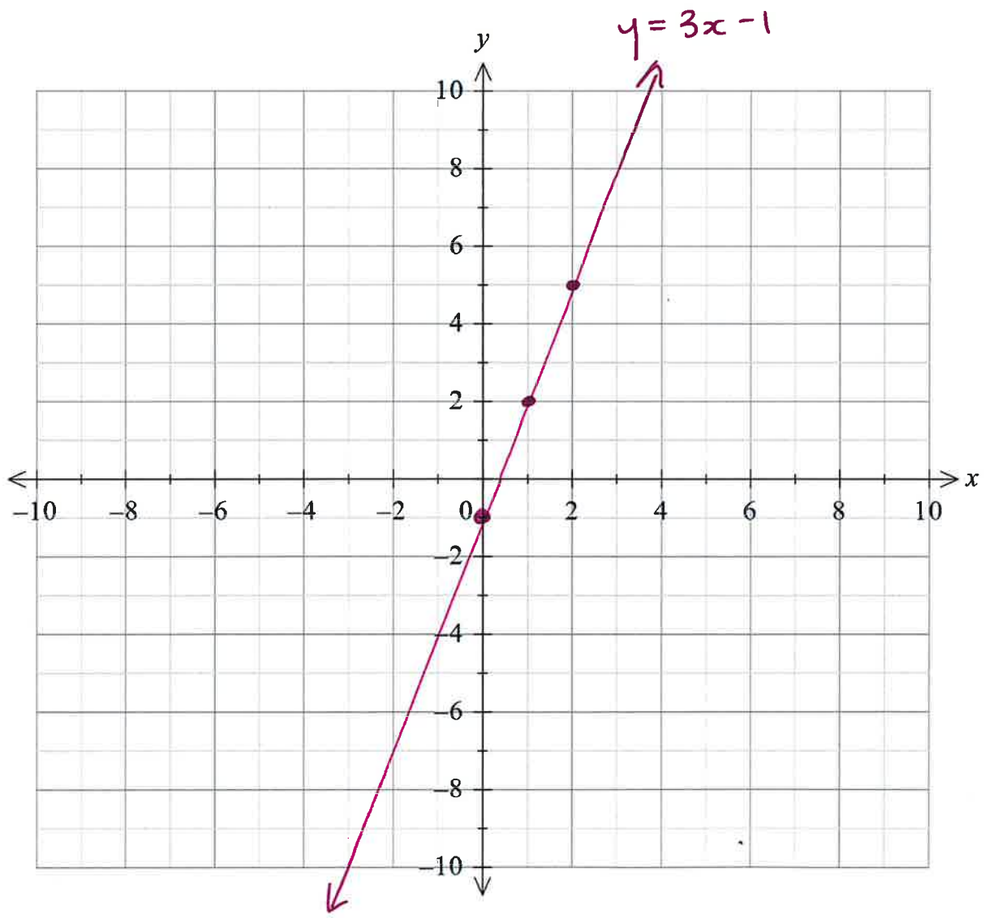
Question 12 (3 marks)

Consider the linear relationship $y = 3x - 1$.

(a) Complete the table of values for $y = 3x - 1$. **1**

x	0	1	2
y	-1	2	5

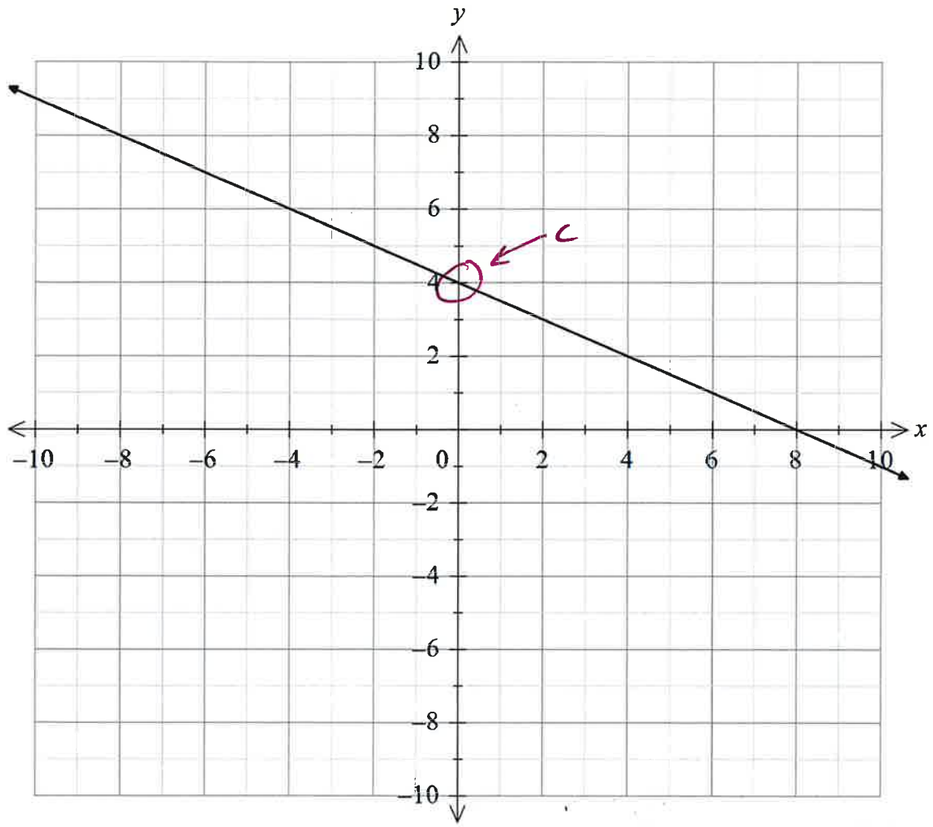
(b) Graph $y = 3x - 1$ on the set of axes below. **2**



/3

Question 13 (2 marks)

Consider the following graph.



Using the gradient-intercept form, or otherwise, find the equation of the line. **2**

$y = mx + c$

$m = \frac{\text{rise}}{\text{run}} = \frac{-4}{8}$ $\quad c = 4$

$= -\frac{1}{2}$

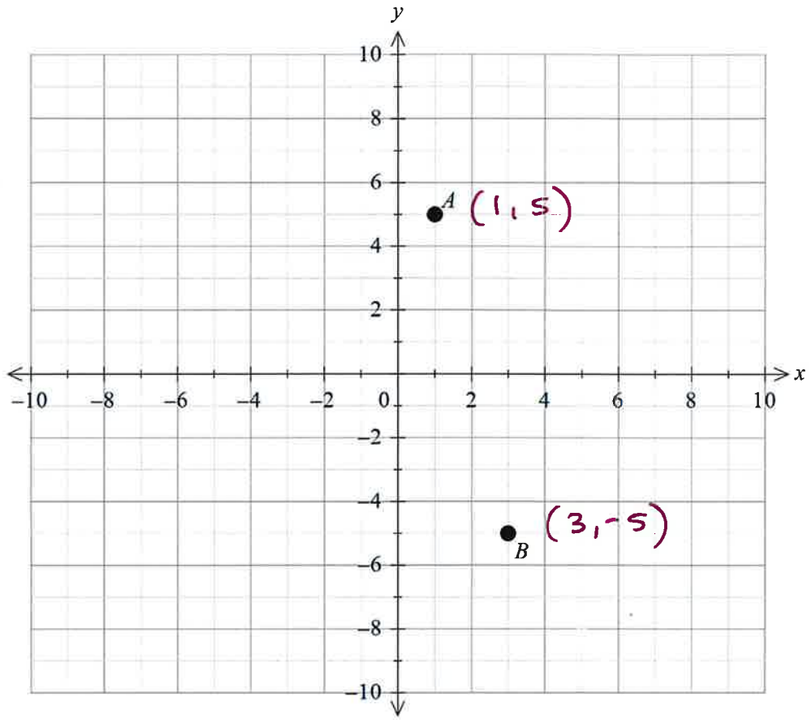
sub $m = -\frac{1}{2}$ & $c = 4$ into $y = mx + c$

$y = -\frac{1}{2}x + 4$

/2

Question 14 (4 marks)

Consider an interval that has endpoints A and B.



(a) Find the midpoint, M, of AB. **2**

$M_{AB} = \left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$

$= \left(\frac{1+3}{2}, \frac{5-5}{2}\right)$

$= (2, 0)$

(b) Find the exact length of the interval AB. **2**

$d_{AB} = \sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$

$= \sqrt{(3-1)^2+(-5-5)^2}$

$= \sqrt{104}$

/4

Question 15 (3 marks)

The average height, H, in centimetres, of a boy between the ages of 5 years and 8 years can be represented by a line with equation $H = 6A + 80$, where A is the age in years.

(a) The gradient of the line is 6. What does the gradient indicate about the height of boys aged 5 to 8? **1**

The height of boys between ages 5 and 8 increases by 6cm a year.

(b) Predict what height, on average, a boy aged 7 years would be. **2**

$H = 6A + 80$

sub in $A = 7$

$H = 6(7) + 80$

$= 122\text{cm}$

∴ a boy aged 7 years old would be 122cm on average.

/3

Question 16 (4 marks)

Consider the following linear relation, $x-2y=8$.

(a) Show that the x-intercept is 8. **1**

let $y=0$

$x-2(0)=8$

$x=8$

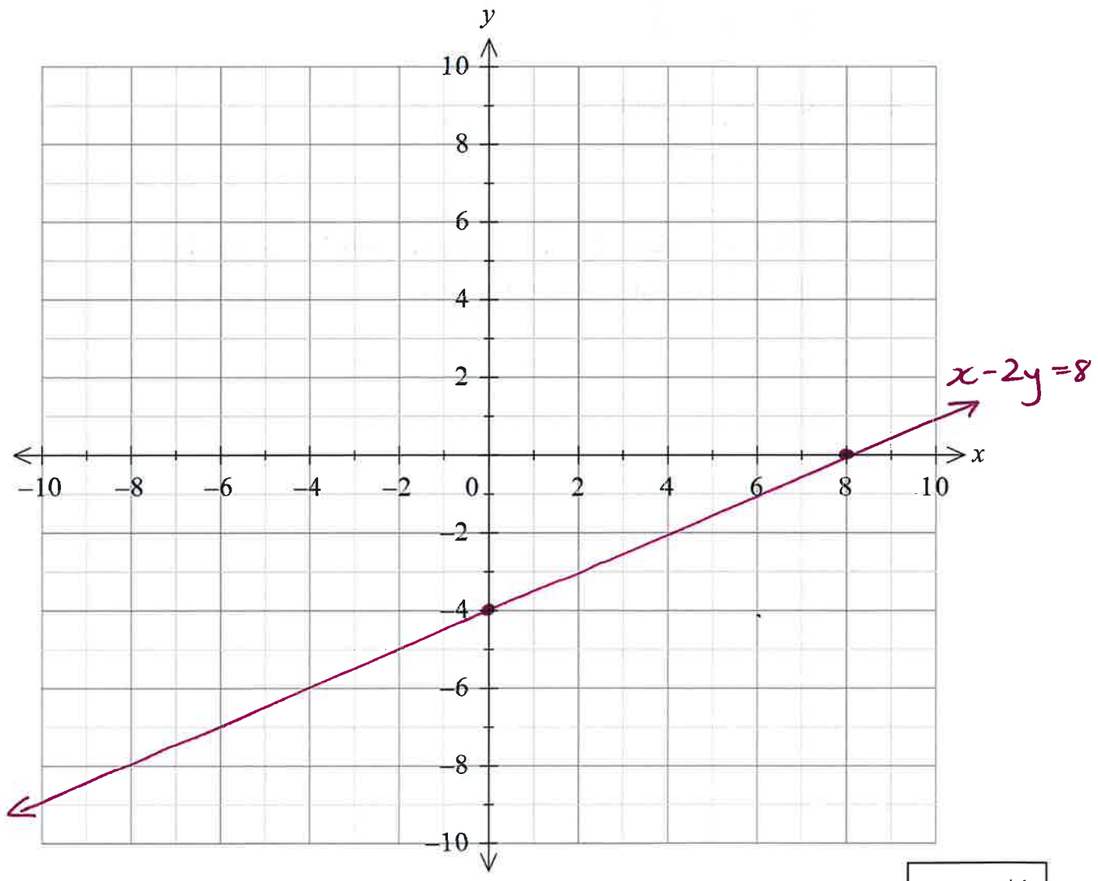
(b) Show that the y-intercept is -4. **1**

let $x=0$

$0-2y=8$

$y=-4$

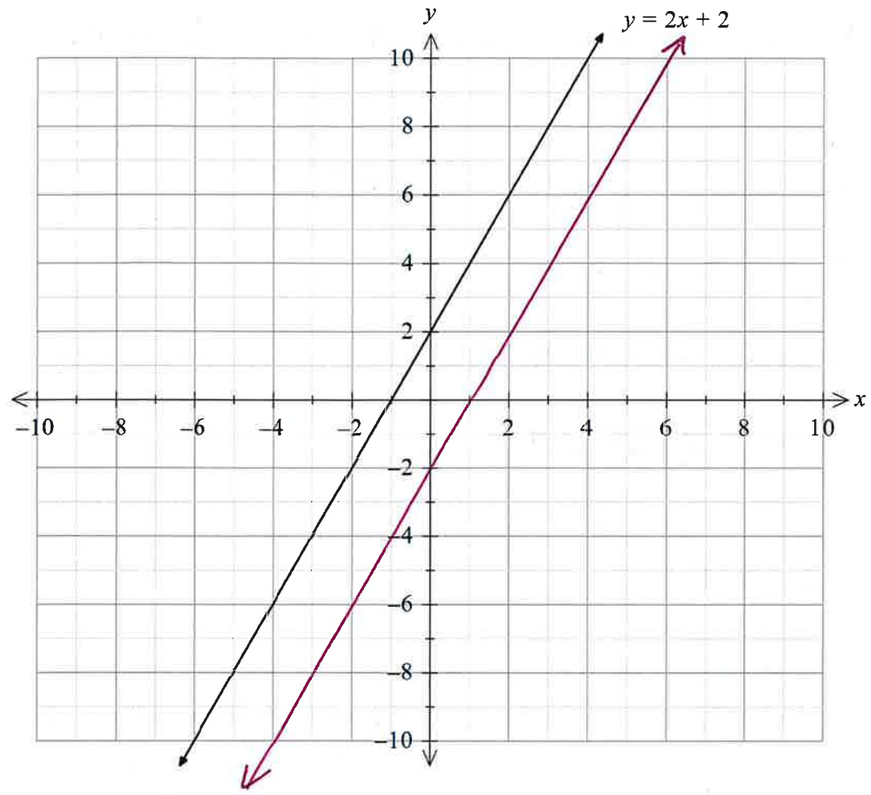
(c) Using the x and y intercepts, plot the linear relation, $x-2y=8$. **2**



/4

Question 17 (4 marks)

Consider the following line $y=2x+2$.



(a) Draw a graph that is parallel to $y=2x+2$ on the same set of axes. **2**

(b) Write the equation of the line you have drawn on the graph. **2**

$y=2x-2$

/4

Question 18 (3 marks)

Find the equation of the line perpendicular to $y=\frac{5}{2}x+3$ that passes through the point $(5,-1)$. **3**

Let m_1 be the gradient of $y=\frac{5}{2}x+3$

$m_1=\frac{5}{2}$

$\Rightarrow m_2=-\frac{2}{5}$ since $m_1 \times m_2 = -1$

$\Rightarrow y=-\frac{2}{5}x+c$

sub in $x=5$ & $y=-1$

$-1=-\frac{2}{5}(5)+c$

$-1=-2+c$

$c=1$

$\therefore y=-\frac{2}{5}x+1$ is the equation of the perpendicular line to $y=\frac{5}{2}x+3$.

END OF PAPER

/3

Booklet B extra writing space

If you use this space, clearly indicate which question you are answering.